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Cell Cycle and Cell Division - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025

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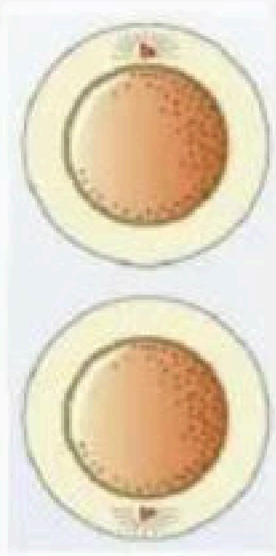
V.V.V.Imp → 20 Mks (SQ)

Good evening
↓
audible? → Ready ⇒

Cell Cycle and Cell Division

BIOLOGYLIVE

1. According to given figure which of the following option is correct : NEET 2025



(1) A – Interphase

(2) A – Telophase

(3) A – Prophase

(4) A – Anaphase

2. Which of the following is "incorrect" statement about meiosis :

(1) One cycle of DNA Replication occure

(2) Two haploid cells are formed at the end of meiosis 4

(3) At the end of meiosis the number of chromosome are reduced to half

(4) Crossing over occur

n

3. An egg cell has "2.5 Pg of DNA" in its nucleus. How much amount of DNA will be in this animal at the "end of G_2 phase of mitosis:"

(1) 2.5 Pg

(2) 5 Pg

(3) 10 Pg

(4) 20 Pg

2n

2n = 5pg

10pg

Original DNA content = 2.5pg → n

After DNA Rep = 2 × 2.5pg = 5pg

end of G_2 phase = 10pg

4. Cells in G_0 phase :

(1) Exit the cell cycle

(2) Terminate the cell cycle

(3) Enter the cell cycle

(4) All of these

5. Given below are two statements :

Statement I : Events of cell cycle are themselves under genetic control

Statement II : Growth and reproduction are characteristic of cells, Indeed of all living organism

Choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

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6. Synapsis occurs between : 2025

(1) Two non-homologous chromosome

(2) mRNA and ribosomes

(3) Spindle fibres and centromere

(4) Two homologous chromosome

Title: Cell Cycle and Cell Division

7. Which of the following statement is correct with respect plant cell:

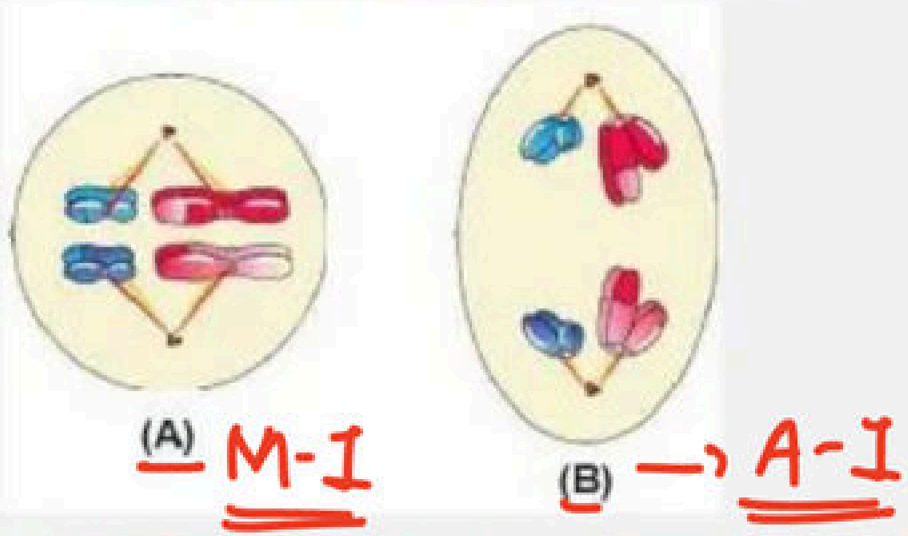
(1) In plant cells cell wall formation start in the center of the cell

(2) Formation of the new cell wall begins with the formation of a simple precursor called cell plate

(3) Cell plate represents the middle lamella are between the walls of two adjacent cells

(4) All of the above

8. Following diagrams A and B showing the phases of a cell division :



Choose the correct option from the following regarding identification of above diagrams:

(1) A = Metaphase - I, B = Anaphase - I

(2) A = Metaphase - II, B = Anaphase - II

(3) A = Metaphase - I, B = Anaphase - II

(4) A = Metaphase - II, B = Anaphase - I

9. How many pollen mother cells should undergo meiotic division to produce 64 pollen grains :

2n

Reductional

(1) 64

(2) 32

(3) 16

(4) 8

$\frac{1}{4} \times \cancel{64}^{16} = \underline{16}$

10. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Meiosis increase the genetic variability in population of organism from one generation to the next

Reason (R) : In meiosis crossing over occur. Crossing over lead to recombination of genetic material on the two chromosome.

In the light of the above statements, choose the correct answer from the options given below

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correction explanation of (A)

→ DNA Rep → Nucleus

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11. Given below are two statements :

Statement I : DNA synthesis occur only during one specific stage in cell cycle. ✓

Statement-II : In animal cells during S phase centriole duplicate in ~~nucleus~~. ✗

→ cytoplasm

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect ✓

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

12. Arrange the following events of meiosis in correct sequence :

(a) Crossing over 2

(b) Synapsis → 1

(c) Terminalisation of chiasmata 3

(d) Disappearance of nucleolus 4

(1) a, b, c, d (2) b, c, d, a

(3) b, a, d, c (4) b, a, c, d ✓

13. Read the following statement and choose correct option :

(a) Crossing over is the exchange of genetic material between " non sister " chromatids of homologous chromosome

(b) Crossing over is the exchange of genetic material between sister chromatids of homologous chromosome ✗

(c) Crossing over is an enzyme mediated process

(d) Enzyme recombinase involved in crossing over

(1) Only a, c correct
(2) Only b, c, d correct
(3) Only a, c, d correct
(4) Only c, d correct

14. Which of the following is correct about G_1 Phase :

(i) Cell is metabolically highly active

(ii) Continuous growth of the cell

(iii) Replicate its DNA

(1) i and ii are correct

(2) Only ii is correct

(3) i, ii and iii are correct

(4) i and iii are correct

15. Match the following :

2025

Column – I	Column – II
a. Metaphase	i. Initiation of condensation of chromosomal material
b. Interphase	ii. Nuclear envelops assembles around the chromosomes clusters
c. Telophase	iii. Chromosomes arranged at equatorial plate
d. Prophase	iv. The period between two cell division

(1) a-i, b-ii, c-iii, d-iv
(2) a-iv, b-i, c-ii, d- iii
(3) a-iv, b-iii, c-i, d-ii
(4) a- iii, b-iv, c-ii, d-i

Title: Cell Cycle and Cell Division

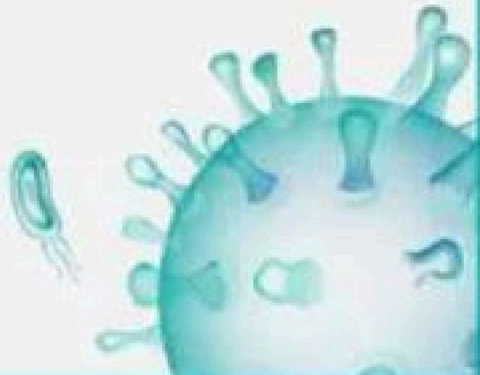
16. In which among the following mitosis division occurs in haploid cells :

(1) Human

(2) Some social insects

(3) Some lower plants

(4) Both (2) and (3)



17. Electron micrographic view of which stage of prophase–I will show clear tetrad : 2025

(1) Leptotene

(2) Pachytene

(3) Zygotene

(4) Diplotene

18. The characteristic feature of diplotene stage of meiosis is recognized by :

(1) Dissolution of synaptonemal complex

(2) Homologous chromosomes of bivalent start separate from each other

(3) X-shaped structure chiasmata formed

(4) All of the above

19. During cell division, the spindle fibres attach to the chromosome at a region called:

(1) Chromocentre

(2) Kinetochore

(3) Centriole

(4) Chromomere

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20. In which among the following stage is marked by terminalisation of chiasmata :

(1) Diakinesis

(2) Diplotene

(3) Pachytene

(4) Zygotene

21. In oocytes of some vertebrates which stage can last for months or years

(1) Diplontene

(2) Pachytene

(3) Diakinesis

(4) Zygotene

22) How many meiotic division would be required to produced 50 synergids in angiosperms

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(1) 100

(2) 50

25 meiotic

(3) 49

(4) 25

Title: Cell Cycle and Cell Division

23. Read the following statement and choose correct answer :

(a) Synapsis of homologous chromosomes takes place during prophase I of meiosis

(b) Division of centromere takes place during Anaphase I of meiosis × 2025

(c) Spindle fibres disappear completely in telophase of mitosis

(d) Nucleoli reappear at telophase I of meiosis

(1) Only a, b correct
(2) Only a, b, c correct
(3) Only a, c, d correct
(4) Only a, b, c, d correct

+ 60 P.G

24. How many meiotic division required to form 60 seeds of higher plants :

(1) 30

(2) 75

(3) 15

(4) 60

pollen + egg

15 + 60 = 75

15 PMC

1 PMC = 4 P.G

15 PMC = 60 P.G

25. Match the following :-

a) Every 24 hours i) Yeast completes its cell cycle

b) 1 hour ii) Typical eukaryotic human cells divide once

c) 95% time iii) Interphase

d) 90 minutes iv) Duration of M-phase in human cell

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii, b-iii, c-iv, d-ii

(3) a-ii, b-iv, c-iii, d-i

(4) a-ii, b-iii, c-iv, d-i

26. Which of the following is not a chracteristic feature during mitosis in somatic cells :

(1) Chromosome movement

(2) Synapsis → meiosis

(3) Spindle fibres

(4) Disappearance of nucleolus.

27. During gamete formation, the enzyme recombinase participates during :

(1) Prophase - II

(2) Metaphase - I

(3) Anaphase - II

(4) Prophase - I

2025

28. The stage during which seperation of the homologous chromosomes take place :

(1) Metaphase - I

(2) Anaphase

(3) Anaphase – I

(4) Zygotene

2025

29. Match the following with respect to meiosis :

(a) Zygotene (i) Terminalization

(b) Pachytene (ii) Chiasmata

(c) Diplotene (iii) Crossing over

(d) Diakinesis (iv) Synapsis

Select the correct option from the following

	(a) (b) (c) (d)		(a) (b) (c) (d)
(1)	(iv) (iii) (ii) (i)	(2)	(i) (ii) (iv) (iii)
(3)	(ii) (iv) (iii) (i)	(4)	(iii) (iv) (i) (ii)

30. Given below are two statements :

Interphase

Statement I : The stage between the "two mitotic division" is called ~~Interkinesis~~

a.I

Statement-II : Homologous chromosome saprate while sister chromatids remain associated at their centromere in ~~anaphase II~~ of meosis II.

Choose the correct answer from the option given below:

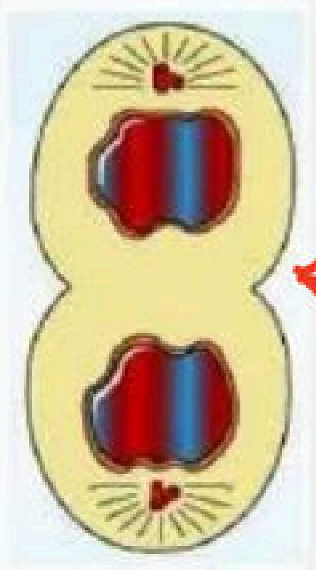
(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

31. Recognise the figure and choose correct option according to figure :



cell furrow

(1) Cytokinesis in plant cell

(2) Cytokinesis in animal cell

(3) Cytokinesis in plant and animal both cell

(4) It is anaphase stage of mitosis.

32. In animal cells, during the S phase, DNA replication begins in the nucleus, and the centriole duplicates in the:

(1) Nucleus

(2) Nucleolus

(3) Both 1 and 2

(4) Cytoplasm

33. During mitosis ER and nucleolus begin to disappear at:

(1) Early prophase

(2) Late anaphase

(3) Early metaphase

(4) Late metaphase

34. Given below are two statements :

reduct

Statement I : In meiosis division number of chromosome in parent and progeny cells same so it is also called ~~equational~~ division. ×

Statement-II : The stage between meiosis I and meiosis II is called interkinesis and is generally short living.

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

35. In some organisms karyokinesis is not followed by cytokinesis as a result of which :

(1) Uninucleate condition arises leading to the formation of syncytium

(2) Multinucleate condition arises leading to the formation of syncytium

(3) Liquid endosperm in coconut.

(4) Both (2) and (3)

2025

36. Match the columns and find out the correct combination:

A.	Doubling of DNA	1. Anaphase
B.	Double the number of chromosomes	2. Cytokinesis
C.	Double the number of cells	3. S-phase
D.	Doubling of cell organelles	4. G_2—phase

(1) A-3 B-1 C-2 D-4
(2) A-1 B-3 C-4 D-2
(3) A-4 B-2 C-1 D-3
(4) A-2 B-4 C-3 D-1

37. Identify wrong pair of statements from the following:

A. Histone synthesis takes place in S phase ✓

B. Doubling of chromosomes occur in S phase of interphase. ✗

C. Nuclei formed after meiosis-I are haploid. ✓

D. Terminalization occurs in Anaphase-I ✗

(1) B and D ✓ (2) C and D

(3) A and B (4) A and D

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38. Synaptonemal complex is a nucleoprotein structure. It is visible or found from

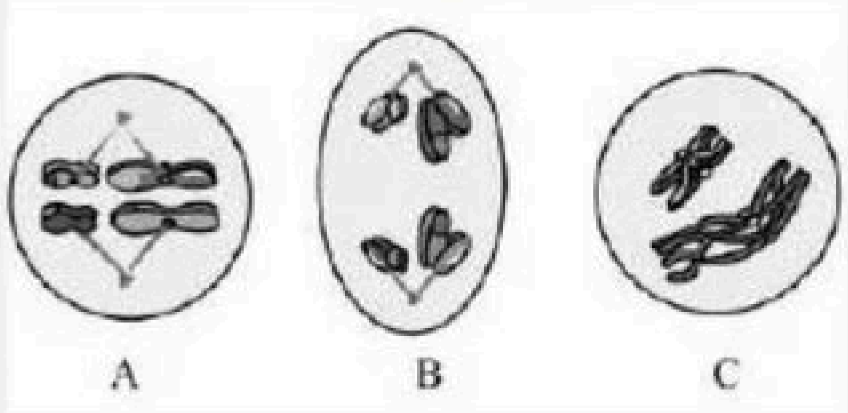
(1) Zygotene through pachytene

(2) Leptotene through diplotene

(3) Zygotene through metaphase

(4) Pachytene through diplotene

39. A, B and C are three stages in the meiosis-I as given in diagrams:



$C \rightarrow A \rightarrow B$

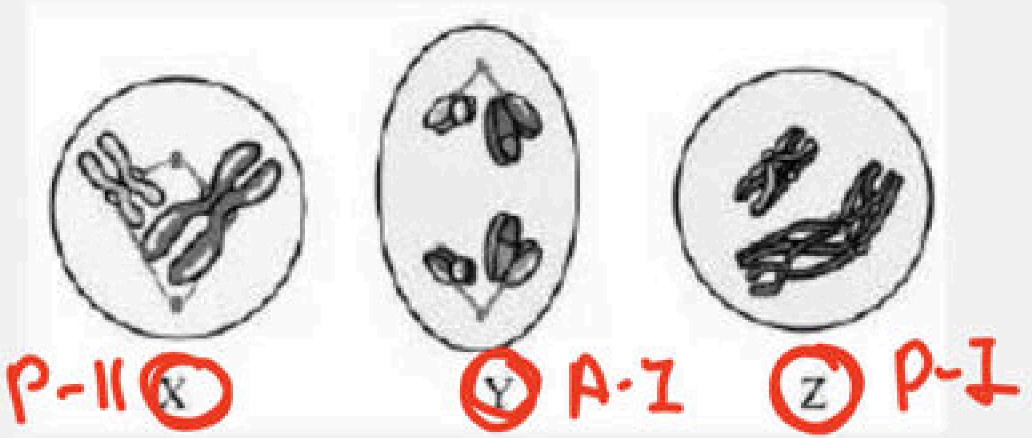
(1) $A \rightarrow C \rightarrow B$

(2) $A \rightarrow B \rightarrow C$

(3) $C \rightarrow A \rightarrow B$

(4) $C \rightarrow B \rightarrow A$

40. X, Y and Z are three stages of cell division in meiosis. Choose the option with their correct identification:



a. Metaphase-II Anaphase-II Prophase-II

b. Prophase-II Anaphase-I Prophase-I

c. Metaphase-I Anaphase-I Anaphase-II

d. Prophase-II Anaphase-II Prophase-1

41. Match the columns and find out the correct combination: 2021

A. Anaphase-I → 2. Separation of two genomes

A. Anaphase-I	1. One spindle apparatus
B. Anaphase-II	2. Separation of two genomes
C. Metaphase-II	3. Two spindle apparatus
D. Telophase-II	4. Separation of two chromatids
	5. Four daughter nuclei

(1) A-2 B-4 C-1 D-3
(2) A-3 B-1 C-4 D-5
(3) A-2 B-4 C-3 D-5 ✓
(4) A-3 B-2 C-5 D-1

42. Pick out the correct statements:

A. Synapsis of homologous chromosomes take place during prophase I of meiosis

B. Division of centromeres takes place during anaphase I of meiosis ×

C. Spindle fibres disappear completely in telophase of mitosis

D. Nucleoli reappear at telophase I of meiosis.

(1) A only (2) C only

(3) A and B only (4) A, C and D only

43. Match the columns and find out the correct combination:

2025

A. Crossing over	1. Diplotene
B. Synapsis	2. Zygotene
C. Weakening of synaptonemal force	3. Leptotene
D. Terminalisation	4. Pachytene
	5. Diakinesis

(1) A-4 B-2 C-1 D-5 (2) A-5 B-3 C-2 D-1

(3) A-4 B-2 C-5 D-3 (4) A-3 B-2 C-4 D-5

2025

44. Match the columns and find out the correct combination:

A. Attraction between homologous chromosomes	1. Crossing over
B. Exchange of genetic material between homologous chromosomes	2. Synapsis
C. Repulsion between homologous chromosomes	3. Segregation
D. Separation of homologous chromosomes	4. Diplotene

(1) A-2 B-1 C-4 D-3 (2) A-1 B-3 C-4 D-2

(3) A-4 B-2 C-1 D-3 (4) A-2 B-4 C-3 D-1

45. Match the columns and find out the correct combination: 2025

A. Metaphase	1. X-shaped structure
B. Diplotene	2. Exchange of part of chromatid
C. Diakinesis	3. Counting of chromosomes
D. Crossing over	4. Terminalisation
	5. Examine the structure of the chromosomes

(1) A-4 B-1 C-2 D-3 ✓(2) A-5 B-1 C-4 D-2

(3) A-2 B-4 C-5 D-1 (4) A-5 B-1 C-3 D-4

46. Which of the following statement(s) is/are correct?

A. The nuclear membrane and nucleolus reappear in telophase I.

B. During metaphase I, the bivalents arrange on the equatorial plate.

C. Metaphase is marked by the alignment of chromosomes at the equatorial plate.

D. During mitosis, the chromosome number of the parent is conserved in the daughter cell.

E. Cell division does not stop with the formation of a mature organism but continues throughout its life cycle.

(1) A, C, and E (2) B only

(3) C and D (4) All of these

47. Choose the correct statement(s) with respect to cell division:

A. The cell duplicates its genome.

B. It synthesizes the other constituents.

C. The events of cell division are under genetic control.

→ nuclear division

D. The M-phase starts with an increase in the number of cell organelles and decondensation of chromosomes.

(1) A & D
(2) C & D
(3) A, B & D
(4) A, B & C

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48. Which one of the following lists contains the correct order of meiotic events?

(1) Separation of sister chromatids, recombination, formation of the synaptonemal complex, separation of homologous chromosomes

(2) Separation of homologous chromosomes, formation of the synaptonemal complex, recombination, separation of sister chromatids

(3) Formation of the synaptonemal complex, recombination, separation of sister chromatids, separation of homologous chromosomes

(4) Formation of the synaptonemal complex, recombination, separation of homologous chromosomes, separation of sister chromatids

v.v.v.Imp

49. Meiosis involves:

(1) Two nuclear divisions and one chromosomal division

(2) One nuclear division and one chromosomal division

(3) One nuclear division and two chromosomal divisions

(4) Two nuclear divisions and two chromosomal divisions

50. In which stage does the synaptonemal complex dissolve?

(1) Zygotene

(2) Pachytene

(3) Diplotene

(4) Diakinesis

51. Which among the following cells are being constantly replaced :

(1) Upper layer of epidermis

(2) Blood cells

(3) Cells of lining of gut

(4) All of these

52. Terminalization, synaptonemal complex, chiasmata, these terms are related to :

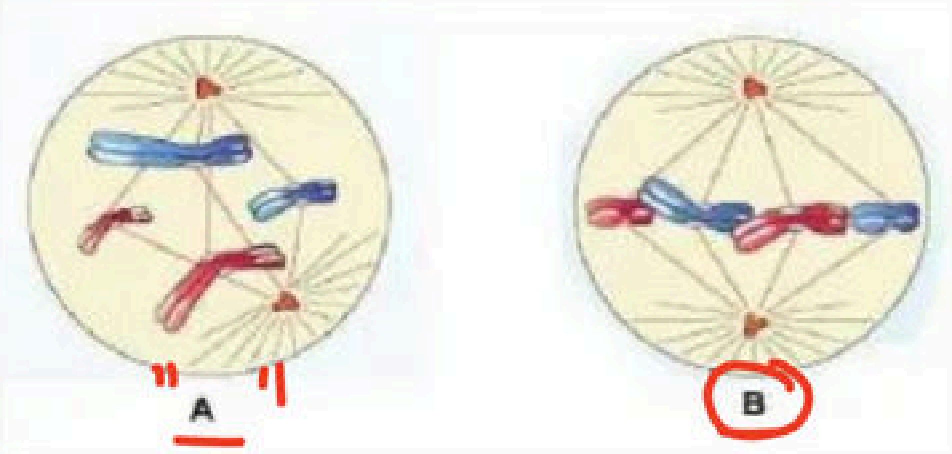
(1) Mitosis

(2) Meiosis I

(3) Meiosis II

(4) All of these

53. Identify the following diagramatic figure and choose correct option :



(1) A – Metaphase II , B – Anaphase II

(2) A – Metaphase I , B – Anaphase I

(3) A – Transition to Anaphase I , B – Prophase

(4) A – Transition to Metaphase , B – Metaphase

54. Recombination is completed by which stage in meiosis:

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) None of these

Title: Cell Cycle and Cell Division

55. How many meiotic division would be required to produced 100 male gametophyte in angiosperms :-

(1) 26

(2) 25

(3) 100

(4) 125

56. Most of the organelle duplication occurs during

(1) G1-phase

(2) S-phase

(3) G2-phase

(4) M-phase

2025

57. Match the following correctly

(a) Chromosomes are moved to spindle equator	i. Anaphase
(b) Centromere splits	ii. Metaphase
(c) Pairing between homologous chromosomes	iii. Pachytene
(d) Crossing over between homologous chromosomes	iv. Zygotene

(1) a-ii, b-i, c-iv, d-iii

(2) a-i, b-ii, c-iii, d-iv

(3) a-iv, b-ii, c-iii, d-i

(4) a-ii, b-iv, c-i, d-iii

58. Crossing over is

(1) Non-enzyme mediated process

(2) Not exchange of genetic material

(3) An enzyme mediated process

(4) None of the above

59. Which of the following events occur during prophase-I in meiosis :

(1) Synapsis

(2) Crossing over

(3) Chiasmata formation

(4) All of the above

60. In a cell amount of DNA is 2C and chromosome no. are 26 these after S-phase amount of DNA and chromosome no. are

4C

26

(1) C & 26 respectively

(2) 3C & 52 respectively

(3) 4C & 52 respectively

(4) 4C & 26 respectively

61. Complete disintegration of nuclear envelope marks the start of :

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

62. The four haploid nuclei found at the end of meiosis differ from one another in their genetic comoposition. Some of this difference is the result of:

(1) Cytokinesis

(2) Respiration of DNA during the "S" phase

(3) Spindle formation

(4) Crossing over during Prophase-I

63. Chromosomes decondense into diffuse chromatin

(1) At the end of telophase

(2) At the beginning of prophase

(3) At the end of interphase

(4) At the end of metaphase

64. Which of the following event marked during mitotic prophase :

(1) Disapperance of the nuclear envelope

(2) Chromosome condensation

(3) Migration of centrosomes towards the cell poles

(4) All of these

65. Given below are two statements :

Statement I : Mitosis usually results in the production of diploid daughter cells

identical

with indentical genetic complement

Statement-II : The growth of multicellular organism is due to mitosis

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

66. This phase of cell cycle is a period of intense synthesis and growth. It constitutes 95% of the duration of cell cycle it is

(1) Interphase

(2) Telophase

(3) Prophase

(4) Anaphase

67. Given below are two statements :

Statement I : Cell plate represents the middle lamella between the walls of two adjacents cells

cytokinesis

Statement-II : At the time of ~~karyokinesis~~, organelles like 2025

mitochondria and plastids get distributed between the daughter cells

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

68. Given below are two statements :

Statement I : Cytokinesis in plant cell takes place by cell-plate formation

Statement-II : Cytokinesis in animal cells by furrowing and is centripetal.

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

Title: Cell Cycle and Cell Division

69. Given below are two statements :

Statement I : Meiosis ensures the production of haploid phase in life cycle of sexually reproducing organism. ✓

Statement-II : Meiosis I is initiated after the parental chromosome have replicated to produce identical ~~chromosome~~ chromatid at S phase. ✗

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect ✓

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

70. A short stage between Meiosis I and Meiosis II is called :

(1) Interphase

(2) Interkinesis

(3) Generation Time

(4) Synthetic phase

71. The smallest stage in M-phase of cell cycle is :

(1) Anaphase

(2) Metaphase

(3) Prophase

(4) Telophase

72. Given below are two statements

Statement I: The process of crossing over takes place in meiosis II

Statement II : The process of crossing over required for variation.

Choose the correct answer from the option given below:

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

73. Which is the correct sequence in cell cycle :

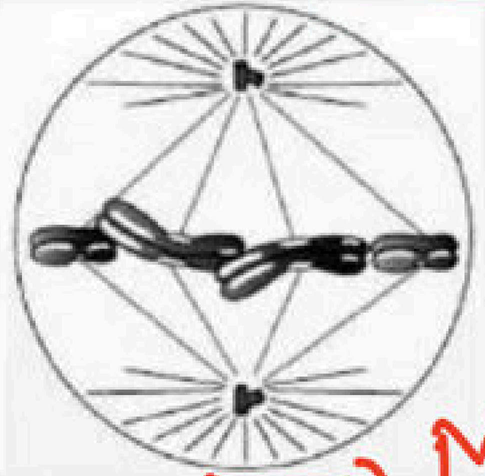
(1) $S - G_1 - G_2 - M$

(2) $S - M - G_1 - G_2$

(3) $G_1 - S - G_2 - M$

(4) $M - G_1 - G_2 - S$

74. Which of the following statement is "incorrect" for the above figure :



Metaphase

→ Meiosis

(1) Formation of synaptonemal complex ✓ ×

(2) Chromosomes are moved to spindle equator and get aligned along anaphase plate through spindle fibres to both poles ✓ → Anaphase ×

(3) Centromeres split and chromatids separate ✓ → Anaphase ×

(4) All of these ✓

75. Some of the cells remain metabolically active, however they do not appear to exhibit proliferation. Which of the following options represent the stage at which cell enters into a non dividing phase :

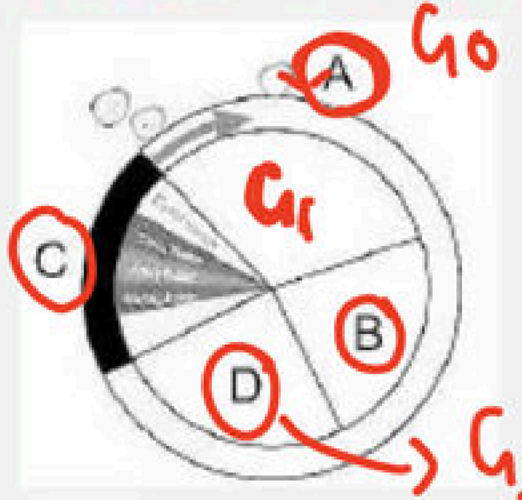
(1) G_0

(2) G_1

(3) G_2

(4) S–phase

76. Which one of the following is correct for the given diagram :



(1) A–Gap 1

(2) C–DNA Synthesis

(3) B–Number of chromosome per cell doubles ✗

(4) D–Protein are synthesised required for mitosis ✓

77. The cell divide to restore the

(1) Mitochondria - cytoplasm ratio

(2) Nucleus - Cytoplasm ratio

(3) Ribosomes - Cytoplasm ratio

(4) ER-cytoplasm ratio

Mitosis

78. Given below are two statements

Statement I: A very significant contribution of meiosis is "cell repair" ✗

Statement II : Mitotic divisions takes place in meristematic tissue of plant. ✓

Choose the correct answer from the option given below:

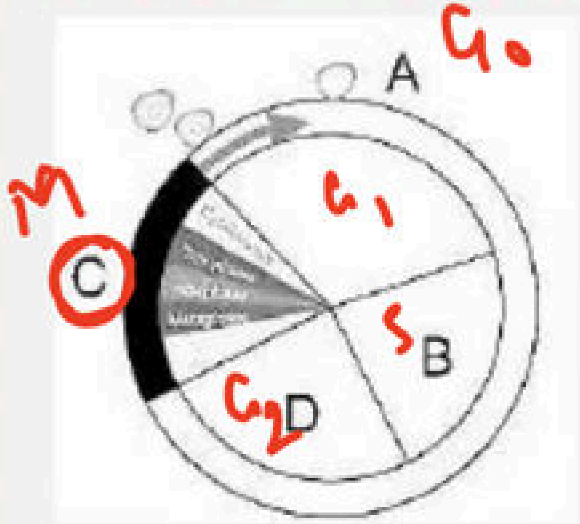
(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

✓(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

79. Which one of the following is correct for the given diagram :



(1) A–Synthetic phase

(2) C–Most dramatic phase

(3) B–Gap 1

(4) D–DNA synthesis

Title: Cell Cycle and Cell Division

80. Given below are two statements :

Statement I : Meiosis-I is ~~equational~~ Red division while meiosis-II is ~~reductional~~ equ division x

Statement II : In Meiosis-I chromosome number bcome half in compare to parental cell while in meiosis II no change in chromosome number in compare to cell formed by meiosis I. ✓

Choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

✓ (3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

81. Diploid chromosome number is 16. What shall be the number of chromatids in each daughter cell after meiosis-I :

(1) 8

(2) 4

(3) 16

(4) 32

10am

9pm

82. In "mitosis" how many time nucleus divide :

(1) Twice

(2) Once

(3) Four time

(4) Does not divide

83. Given below are two statements :

Statement I : Prophase II is much simpler than prophase I

Statement II : In metaphase I bivalent chromosome align on the equatorial plate.

Choose the correct answer from the options given below.

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both, Statement I and Statement II are correct

84. Which of the two events restore the normal number of chromosomes in the life cycle?

(1) Mitosis and Meiosis

(2) Meiosis and Fertilisation

(3) Fertilisation and Mitosis x

(4) Only Meiosis

85. If the number of bivalents is 8 in metaphase-I, what will be the number of chromosomes in daughter cells after meiosis-I and meiosis-II, respectively?

(1) 8 and 4

(2) 4 and 4

(3) 8 and 8

(4) 16 and 8

86. At what phase of meiosis are there two cells, each with sister chromatids aligned at the spindle equator?

(1) Anaphase II

(2) Metaphase II

(3) Metaphase I

(4) Anaphase I

87. During which stages (or prophase I substages) of meiosis do you expect to find the bivalents and DNA replication, respectively?

(1) Pachytene and interphase (between two meiotic divisions)

(2) Pachytene and interphase (just prior to prophase I)

(3) Pachytene and S phase (of interphase just prior to prophase I)

(4) Zygotene and S phase (of interphase prior to prophase I)

88. During Anaphase-I of meiosis:

(1) Homologous chromosomes separate

(2) Non-homologous chromosomes separate

(3) Sister chromatids separate

(4) Non-sister chromatids separate

89. A bivalent of "meiosis I" consists of: 2025

(1) Four chromatids and two centromeres

(2) Two chromatids and one centromere

(3) Two chromatids and two centromeres

(4) Four chromatids and four centromeres

90. During mitosis, after the separation of centromeres, the chromatids move toward the opposite poles of the spindle. What term is used for these chromatids?

(1) Daughter chromosomes

(2) Kinetochores

(3) Half spindles

(4) Centrosomes

91. Which of the following proteinaceous components of the cell cytoplasm help in the initiation of the assembly of the mitotic spindle?

(1) Microtubules
(2) Microbodies
(3) Centromeres
(4) Kinetochores

Title: Cell Cycle and Cell Division

92. The M phase starts with the nuclear division, corresponding to the separation of daughter chromosomes called ...A... and usually ends with the division of cytoplasm, called ...B....

(1) A - Cytokinesis, B – Karyokinesis

(2) A - Interkinesis, B - Cytokinesis

(3) A - Karyokinesis, B – Cytokinesis

(4) A - Interkinesis, B - Karyokinesis

meiosis haploid

93. 'X' ensures the production of 'Y' phase in the life cycle of sexually reproducing organisms, whereas fertilization restores the 'Z' phase. Identify X, Y, and Z.

diploid

(1) X - Mitosis, Y - haploid, Z - haploid

(2) X - Mitosis, Y - diploid, Z - diploid

(3) X - Meiosis, Y - haploid, Z – diploid

(4) X - Meiosis, Y - diploid, Z - diploid

94. Meiotic cell division is also termed as reduction division because:

(1) Involvement of gametes

(2) Doubling of chromosomes

(3) Elimination of chromosomes

(4) Number of chromosomes becomes halved

95. The homologous chromosomes separate, while sister chromatids remain associated at their centromeres during:

(1) Metaphase-I of meiosis

(2) Anaphase-I of meiosis

(3) Metaphase of mitosis

(4) Anaphase of mitosis

96. Which of the following does not occur in Anaphase I?

(1) Segregation of homologous chromosomes

(2) Contraction in spindle

(3) Poleward movement of chromosomes

(4) Division of centromere

97. During the pachytene stage of meiosis, the chromosomes appear:

(1) Single stranded

(2) Two stranded

(3) Six stranded

(4) Eight stranded

98. Among the following, which one is the longest phase in prophase of meiosis?

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

99. After meiosis-I, the two chromatids of a chromosome are:

(1) Genetically similar

(2) Genetically different

(3) There occurs only one chromatid in each chromosome

(4) None of the above

100. Which is unique to mitosis and does not occur in meiosis?

(1) Homologous chromosomes cross over

(2) Homologous chromosomes pair and form bivalents

(3) Homologous chromosomes behave independently only in mitosis

(4) Chromatids are separated during anaphase

THANK YOU

BIOLOGYLIVE

• Tom:- Biomolecule

IMP